

Vedant Sarvesh Ranade

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EDUCATION

Northeastern University, Khoury College Of Computer Science

Boston, MA

Pursuing Master of Science in Artificial Intelligence, Concentration in Machine Learning

Sep. 2025 – May.2027

- Coursework: Linear Algebra and Probability, Foundations of AI and Algorithms

University of Mumbai

Mumbai, India

Bachelor's in Artificial Intelligence and Data Science; GPA 3.74/4.00

Nov. 2021 – May 2025

- Coursework: Deep Learning, Reinforcement Learning, Statistics, Cloud Computing, and Big Data Analytics

TECHNICAL SKILLS

Languages: Python, JavaScript, C, SQL(Postgres), MongoDB

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face

Deep Learning: CNN, Sequence Modeling, Timeseries Forecasting, Generative Models, Large Language Models, Nbeats

Natural Language Processing : BERT, CLIP, TF-IDF, NER

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Feature Engineering

Deployment: Azure, AWS, Streamlit, Tensorboard

Other Frameworks: ReactJS, FastAPI, Flask, Node.js, WordPress

Other Developer Tools: Git, Google Cloud Platform, Microsoft Office, Cursor, Firestore

PROJECTS

EduBoost: AI Based Educational Tools | *Tensorflow, ReactJS, Azure, FastAPI*

Sept. 2024 – May 2025

- Led AI research and model fine-tuning to enhance the performance of various existing supervised models.
- Developed a sketch-based math solver (like iOS 18) using CNNs capable of interpreting handwritten equations.
- Fine-tuned the Caffe model for illustration colorization, to achieve smoother and more realistic gradients.
- Integrated Transformer models BLIP and CLIP to generate an illustration summary.
- Designed a full-stack solution with ReactJS frontend, Firebase authentication, and MongoDB backend.
- Deployed all AI models on Azure, ensuring scalability and high availability.

Deepfake Image Detection | *Python, Flask, Azure, Tensorflow, PIL*

Jan 2024 – May. 2024

- Architected a custom CNN model achieving 92% accuracy in detecting deepfake images.
- Built and deployed a Flask-based web application for real-time deepfake verification.
- Designed and deployed an API for seamless integration of detection services.
- Deployed the system on an Azure Virtual Machine, ensuring high-performance cloud execution.

AI Based Navigation System for Drones | *Python, Unity, TCP Sockets, Flask*

Sept. 2023 – Dec 2023

- Engineered an AI-driven drone navigation system in Unity, with Python-based control.
- Evaluated and benchmarked multiple AI algorithms across key performance metrics to ensure robustness.
- Established real-time communication between the Unity simulation and Python backend using TCP sockets.
- Developed a Flask-based dashboard with automated SMTP notifications for a real-time Parcel Delivery app.

Demand Based Dynamic Pricing Using AI | *Python, Scikit-Learn, Streamlit*

Jan. 2024 – May 2024

- Developed a high-dimensional ensemble model using Scikit-Learn for demand forecasting and price optimization.
- Integrated real-time data of weather and temporal APIs on location to dynamically adjust pricing strategies.
- Automated pricing, invoicing, with UPI through a Streamlit dashboard, streamlining the workflow.

EXPERIENCE

Python Developer Intern

June 2025 – July 2025

Webcreations, Dombivli, India

- Upgraded management portal using Flask and SQL for real-time data visualization.
- Implemented a 3D rendering model using transformers, solving a complex technical requirement for the client.
- Optimized workflows for large data directories, proposing architectural upgrades improving system scalability.

Web Developer Intern

Dec. 2023

Webcreations Dombivli, India

- Developed and maintained a PHP-based backend automation and ensured efficient data management using SQL.
- Customized websites as per evolving requirements, improving user experience, and demonstrating adaptability.
- Collaborated cross-functionally with design and development teams to accelerate project delivery and quality.